

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation followed in Python, HTML, CSS, and JavaScript files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, model files, dataset, and Flask application.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as Cloud_Cover, Annual, prediction, and scaler are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like home(), predict_page(), and predict() clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are included to explain important sections of the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	Flask routes and machine learning components are separated into logical sections.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary and duplicate code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation is present, but advanced exception handling can be improved in future versions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles routing and prediction requests	High

2	Prediction Function	Python	Used to process user input and generate flood predictions	High
3	HTML Templates	HTML	Shared across different web pages	High
4	CSS & JavaScript	CSS, JavaScript	Used throughout the web application for styling and interaction	High
5	Trained Model & Scaler	XGBoost, Joblib	Reused whenever a prediction request is made	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project files are well organized and easy to navigate.
Readability	5	Meaningful variable names, indentation, and comments improve readability.
Reusability	5	Components such as the prediction model, Flask routes, and templates can be reused in future projects.
Documentation / Comments	4	Comments explain the main logic, though additional documentation could further improve maintainability.
Overall Score	4.8/5	The project follows good coding practices with a clean and maintainable structure.